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Backend, API Validation and Application Delivery Portfolio

Master of Computer Science candidate at the University of Sydney, focused on backend/API systems, application delivery, release readiness, deployment checks, test evidence and automation-assisted implementation.

Portfolio focus: work samples for graduate software roles involving backend services, API contracts, UAT, release gates, deployment support, runtime troubleshooting, runbook writing and measurable system validation.

Selected Project Evidence

FinX WeChat Mini Program Event Platform

Application Delivery and API/UAT Support Intern | WeChat Mini Program + Django/DRF API

- Delivered a Django 5.2 + DRF backend (139 URL routes, 47 test files) and native WeChat Mini Program (45 page directories/subpackages) for authentication, event publishing, ticketing, orders, IM, check-in, invoices, refunds and client error reporting.
- Validated API behavior through targeted checks for health, SMS/token auth, idempotent order creation, WeChat Pay prepay/notify/query, COS uploads and IM UserSig generation.
- Built release-readiness gates covering Django checks, migration drift checks, ruff, pytest, JSON validation, i18n/WXML coverage, miniapp unit tests and manual smoke-test paths.
- Prepared Tencent Cloud CVM deployment assets with Nginx reverse proxy, Gunicorn bound to localhost, systemd service management, HTTPS routing, health probes and rollback notes.

Django/DRF

API/UAT

WeChat Pay

Nginx/Gunicorn/systemd

Tencent Cloud

USYD Capstone - AI Hub PII Redaction API

Graduation capstone project | FastAPI service | API contract | OCR/file ingestion | benchmark reporting

- Built and documented a callable FastAPI redaction service with `/healthz`, `/readyz`, `/version`, `/v1/redact`, `/v1/redact-file`, OpenAPI docs and stable JSON span output.
- Extended the service from text redaction to multi-format file ingestion for TXT, DOCX, PDF and image inputs with OCR/parser metadata.
- Contributed reliability gates around pytest, ruff, CI checks, Docker/arm64 build review and deployment readiness for AI Hub hardware accessed through SSH/Tailscale.
- Consolidated benchmark reporting across a 14,000-document holdout, where the fine-tuned/distilled LLM student model reached precision 0.986, recall 0.970 and F1 0.978.
- Tracked deployability evidence across high-risk recall, parse rate, over/under-redaction cost, latency, throughput and robustness testing.

FastAPI

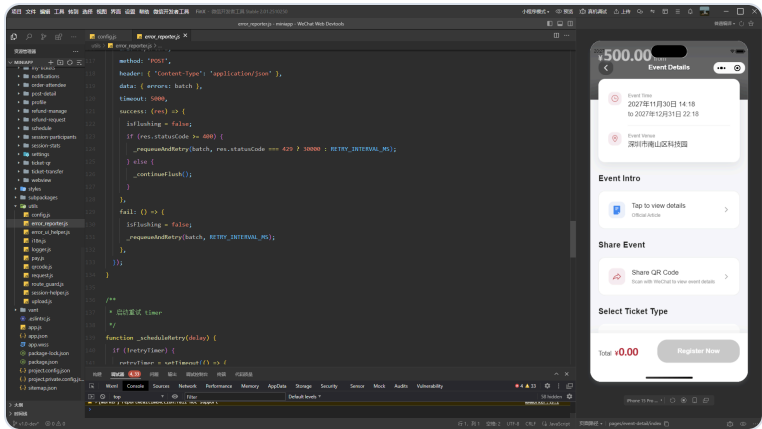
OpenAPI

OCR/file ingestion

pytest/ruff

Benchmarking

Runtime and Automation Proof



FinX - event flow running in WeChat Developer Tools

Local WeChat Developer Tools view of the FinX Mini Program event-detail flow, with the project imported from `miniapp/` and the backend/API surface validated during portfolio preparation.

- Shows an end-to-end user journey: event details, event time and venue, official article entry, QR-code sharing and ticket selection.
- DevTools frame also exposes the frontend route tree and client error reporter, making the screenshot useful as engineering evidence rather than only UI evidence.
- Current repo checks include miniapp unit tests, i18n key validation, Django system checks, migration drift check and targeted API regression suites.
- Useful for demonstrating application delivery across frontend, backend API, UAT, release gates and runbook-oriented handover.

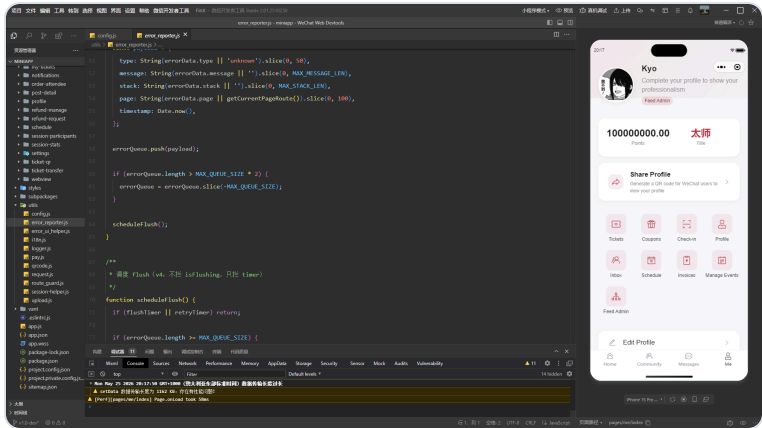
WeChat DevTools

Mini Program

Django/DRF API

UAT gates

Local backend



FinX - role dashboard, operations entry points and permission surface

DevTools view of the user dashboard after sign-in, showing profile state, role badge and operational entry points such as tickets, check-in, invoices, event management and feed administration.

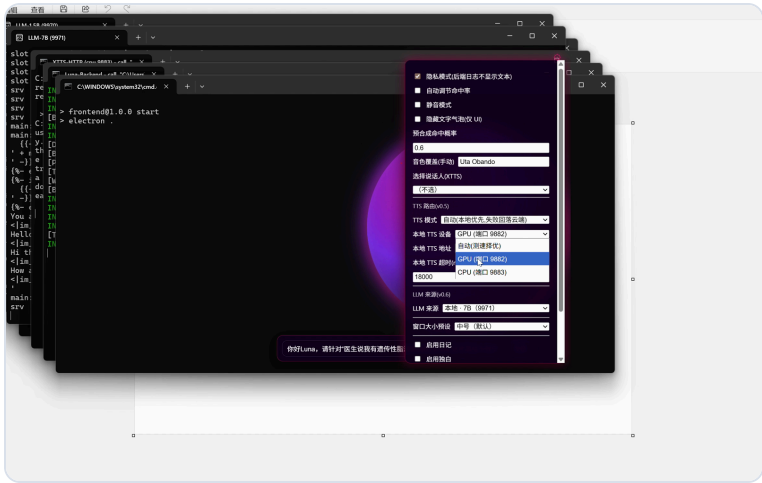
- Demonstrates that FinX is more than a landing page: the product has multiple authenticated modules and role-dependent operations.
- The visible Feed Admin and Manage Events entries align with the backend permission and current-user payload work used by the miniapp.
- Covers role-based permissions, multi-module operations and authenticated UX surface.

Role surface

Permissions

Authenticated UX

Operational modules



Luna - local AI desktop agent with service orchestration

Built a local-first Electron + FastAPI AI desktop agent that orchestrates local Qwen LLM servers, XTTS speech services, cloud fallback paths, health checks, readiness polling, privacy controls and portable data storage.

- Orchestrated a multi-process Windows runtime through a single launch script: Qwen2 1.5B/7B `llama.cpp` servers, XTTS GPU/CPU HTTP services, FastAPI backend and Electron UI.
- Implemented reliability controls for AI service degradation, including local/cloud LLM routing, local/cloud/auto TTS routing, request timeouts, startup warmup and fallback audio behavior.
- Added privacy and portability safeguards through `.portable`/`LUNA\_PORTABLE` data-root isolation, ignored private runtime data, privacy-mode logging and local-only routing options.
- Wrote runbook-style notes covering endpoint contracts, startup order, local model ports, routing modes and common failure checks.

FastAPI

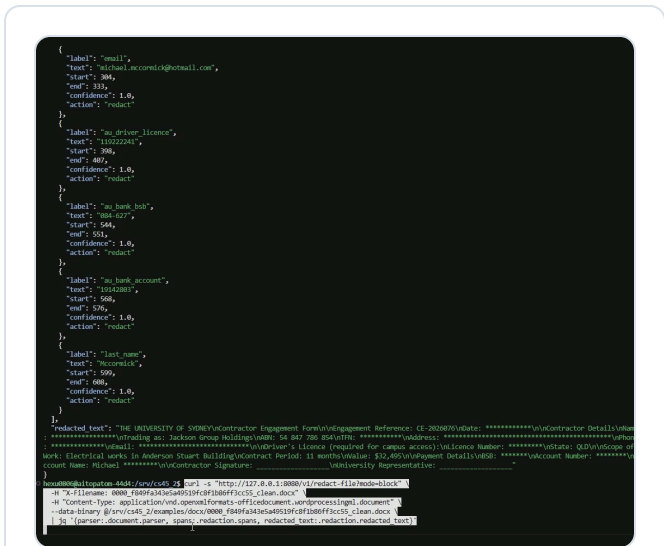
Electron

Local LLM/TTS

Health/readiness

Fallback routing

Graduation Capstone API Demo Evidence



DOCX ingestion and redaction API

Terminal demo of `/v1/redact-file` returning detected spans and redacted text from a DOCX input,

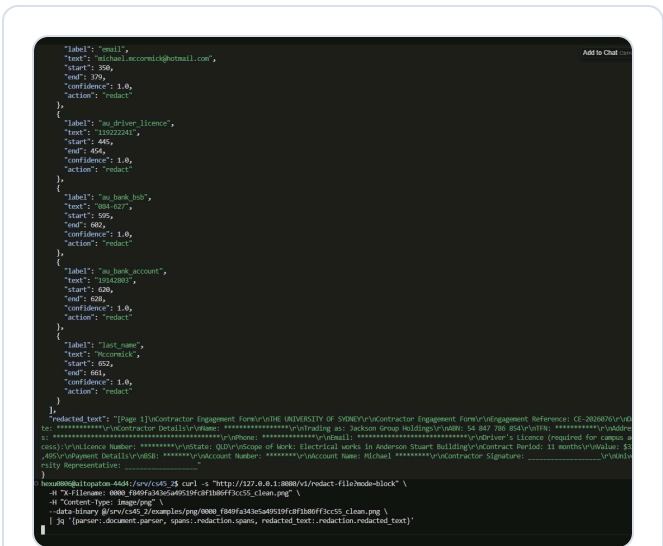


Image OCR ingestion and redaction API

Terminal demo of image upload through OCR/parser flow, with structured redaction spans

showing observable extraction and span-alignment output.

for email, licence, bank BSB, account number and name entities.

The Capstone repository is private; this portfolio uses local demo frames, API contract notes and benchmark/deployment summaries rather than a public repository link.

Earlier Technical and Research Experience

China Eastern Technology Application R&D Center
Technical Intern | regulated maintenance workflow exposure

Built exposure to A320/B737 structure and systems training, ERJ-145 maintenance support, and MD-11 materials record updates.

Regulated workflows

Technical records

Operations exposure

Research and Thesis Work
Outstanding thesis recognition | RPM-driven deep learning fault diagnosis

Built an RPM-based image-representation workflow for vibration-signal fault diagnosis experiments, with model comparison and technical reporting; also explored KAN/PySR interpretability as a separate research extension.

RPM features

Signal data

Model evaluation

MATLAB

Interpretability

Backend, API and Delivery Skills

Service Validation
REST/OpenAPI review, Postman/httpx checks, pytest basics, acceptance criteria, regression checks, issue reproduction notes and release readiness checklists.

Runtime and Deployment
Linux logs, Nginx/Gunicorn/systemd verification, Docker review, Tencent Cloud CVM, local service startup orchestration and health-check documentation.

Current Development Focus

Currently deepening Linux/networking fundamentals, Python tooling, API troubleshooting, SQL basics, and data-structure/algorithm readiness as part of graduate-role preparation.

This portfolio is a concise work-sample attachment prepared for graduate engineering applications. It summarizes project evidence; source access or fuller walkthroughs can be discussed during interviews where appropriate.